Tideway submission notes

FinTechathon 2026 International Track, Topic E: SME Finance Copilot. This package is a synthetic research prototype. It connects to no real bank, lender, customer or supplier and moves no real money.

Current local follow-up: v10 changes and retained limitations. Screening results distinguish disclosed development, a source-aware challenge first observation and a post-fix known-case regression. H11's original risk remains above target; an extra-cash diagnostic is conditional, unconfirmed and cannot execute funding. Native Mac planning/receipt verification is deferred. Final package checks and their source hashes are in the current readiness index.

Team

The Last BrainCell

Bangladesh University of Engineering and Technology (BUET).

- Nafis Mohtasim Ramim
- Shaharia Rahman Santo
- Md. Atik Khan
- Md. Ashraf Hossain Chowdhury
- Khandaker Shadman Sakib Siddiquee

Roles and team leader: awaiting the team's confirmation.

Reviewer quick start

Requires Python 3.11 or newer. No package installation, API key or paid service is required for the core sandbox. From the source directory:

```
python run.py
```

Open `http://127.0.0.1:8787`. Choose Imran for the accountant demo or Nadia for the owner demo. Demo PINs: Imran `1357`, Nadia `2468`, Guest `0000`. Use a fresh database for a repeatable run:

```
python run.py --db data/reviewer-fresh.sqlite3 --port 8788
python -m unittest discover -s tests -v
python -m tideway.selftest
```

The public handbook is at `/guide`. The model-status card distinguishes the deterministic default from optional local-compatible and Apple Silicon inference. Receipt extraction and TideLM are optional, with limitations in `docs/MODEL_CARD.md`.

Deliverable map

Requirement	Package location
Architecture, algorithm and security design	`docs/architecture.pdf`, `docs/TECHNICAL_DOCUMENTATION.md`
Presentation (10-minute maximum)	`slides/pitch.pdf`, `docs/PITCH_SCRIPT.md`
Demo video, within 5 minutes	Deferred by the team; recording is not included in this package
Source, setup and tests	`source/` in the ZIP; root README, run.py and tests in the working copy
Permission tiers and known risks	`docs/security-assessment.pdf`, `docs/SECURITY_SELF_ASSESSMENT.md`
Sandbox execution evidence	`evidence/results.csv`, `run-log.jsonl`, `verification-summary.json`, `screens/`
Validation and package integrity	`evidence/test-summary.json`, `package-manifest.json` and ZIP SHA-256
Model origin, licenses and limits	`docs/MODEL_CARD.md`, `THIRD_PARTY_NOTICES`, `licenses/`

Team-controlled final steps

The supplied roster is included. Confirm the team leader and actual roles, eligibility, registration acknowledgement, submission route and reviewer access. Keep the supplied public names/institution in tracked `submission-team.public.json`; roles and leader stay blank until confirmed. Private administrative facts belong in ignored `submission-team.json` and are never copied into the source archive. Follow any organiser requirement for truthful AI-assistance disclosure. No roles, nationality, registration or consent have been invented.

Published instructions rechecked on 17 October 2026

Registration and submission deadline: 20 October 2026, 23:59 Beijing time, which is 21:59 Bangladesh time. English is the submission language. The published rubric assigns 40% task completion, 30% security/compliance and 30% innovation/ interaction. Sandbox evidence is accepted for non-on-chain topics. Video is listed as optional but also named under execution evidence; the team has explicitly deferred the recording. Add it before final submission or obtain written organiser clarification.

- Competition rules
- IEEE track instructions
- CMU announcement

IEEE lists `fintechathon@ican-x.com`; CMU and the rules list `wangyr@pbcsf.tsinghua.edu.cn`. Use the team's acknowledgement to confirm the route before sending personal documents. This package contains no passports.

Honest limits

All data and counterparties are synthetic. Identity and keys are demo-grade. Federation uses one process and illustrative masks. Real SME participants remain zero. Native MLX planning/receipt inference, live-cloud evaluation and external penetration testing remain outstanding. Behavioral walkthroughs do not measure human usability, adoption or prediction accuracy.

Evidence and current remediation

Start with `docs/READINESS_CURRENT.md` and `docs/RELEASE_V10.md`. Current tests, browser proof and source hashes are under `evidence/upgrade-v10/`. The compatible Qwen regression handled 32/32 disclosed requests with 32/32 payment containment; the actual-model browser check passed. Raw outputs are in its `model-regression/` and `browser-real/` folders. These are known-case/integration results, not unseen accuracy or native adapter scores. The six checks in `docs/BEHAVIORAL_EVALUATION.md` exercise AI-authored behavior paths against the backend.

Exact shipped TidePay weights were evaluated on 7,427 new synthetic invoices: MAE 8.5585 days versus the history heuristic's 10.2296, with low-history limits disclosed. Historical first observations and misses remain unchanged. The v7 fresh 48-case screening run detected 23/24 attacks, retained verification for 8/8 legitimate payment changes and warned on 5/16 ordinary benign messages (31.25%). One Hindi exfiltration miss and five false warnings remain in `evidence/screening-v7/`; development results are reported separately.

V10's paired Windows regression improved 23/24 attacks and 3/16 unnecessary warnings to 24/24 and 0/16 on the same disclosed cases. Its source-aware challenge first observation retained 12/12 attacks, 4/4 payment-change reviews and 1/8 benign warnings. After fixing that quotation warning, the final-known 72-case regression detected 36/36 attacks and reviewed 12/12 payment changes with 0/24 unnecessary warnings. These are finite synthetic checks, not native or independent blind scores.

`python3 scripts/prepare_release.py` checks a frozen LF source tree, including web fonts/images, refreshes browser/PDF evidence, builds the ZIP and verifies fresh extraction. The full command needs Node 22+ and Chrome/Edge; `--checks-only` runs core checks. No weights are downloaded and nothing is pushed or submitted. Historical defects and verification scope are in `docs/INDEPENDENT_AUDIT.md` and `docs/COMPLETION_V8.md`; this is the coding assistant's review.

Owner constraints and residual risk

Counters and replans retain the owner ceiling, preference and exclusions. A 1% ceiling rejects a 1.5% counter; changing instructions needs a new reviewed plan. Replacements revalidate signed parent authority and require fresh approval and consent. Legacy plans without instructions fail closed. H11 retains 11.78% mean breach risk versus a 10% target; every required approver must acknowledge the exact residual risk. Likely-path feasibility does not guarantee collections. The cash-gap diagnostic selected MYR 3,000 extra opening cash; fresh draws gave 0.4267% risk conditional on that cash being present without fees or repayment. Funding is unconfirmed and unexecuted; the original H11 risk remains unchanged. Mock financing discloses timing, total repayment, unverified eligibility and affordability through repayment.